contaminating any attempt to simultaneously ascertain its velocity. Although, as we have seen, quantum uncertainty is more general than the "disturbance" explanation indicates, Einstein, Podolsky, and Rosen invented what appeared to be a convincing and clever end run around *any* source of uncertainty. What if, they suggested, you could perform an indirect measurement of both the position and the velocity of a particle in a manner that never brings you into contact with the particle itself? For instance, using a classical analogy, imagine that Rod and Todd Flanders decide to do some lone wandering in Springfield's newly formed Nuclear Desert. They start back to back in the desert's center and agree to walk straight ahead, in opposite directions, at exactly the same prearranged speed. Imagine further that, nine hours later, their father, Ned, returning from his trek up Mount Springfield, catches sight of Rod, runs to him, and desperately asks about Todd's whereabouts. Well, by that point, Todd is far away, but by questioning and observing Rod, Ned can nevertheless learn much about Todd. If Rod is exactly 45 miles due east of the starting location, Todd must be exactly 45 miles due west of the starting location. If Rod is walking at exactly 5 miles per hour due east, Todd must be walking at exactly 5 miles per hour due west. So even though Todd is some 90 miles away, Ned can determine his position and speed, albeit indirectly.

Einstein and his colleagues applied a similar strategy to the quantum domain. There are well-known physical processes whereby two particles emerge from a common location with properties that are related in somewhat the same way as the motion of Rod and Todd. For example, if an initial single particle should disintegrate into two particles of equal mass that fly off "back-to-back" (like an explosive shooting off two chunks in opposite directions), something that is common in the realm of subatomic particle physics, the velocities of the two constituents will be equal and opposite. Moreover, the positions of the two constituent particles will also be closely related, and for simplicity the particles can be thought of as always being equidistant from their common origin.

An important distinction between the classical example involving Rod and Todd, and the quantum description of the two particles, is that although we can say with certainty that there is a definite relationship between the speeds of the two particles—if one were measured and found to be moving to the left at a given speed, then the other would